

SOA Project Review-2

Smart Item Recovery Portal AI Driven Smart Matching Management System

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Problem Statement

- **Traditional lost and found systems mainly depend on manual reporting, record keeping, and searching, which makes the process time-consuming, inefficient, and difficult to manage as the number of users and items increases.**
- **Users often have to manually compare the descriptions of lost and found items, while the absence of an intelligent matching mechanism can result in potential matches being missed.**
- **In addition, conventional systems may provide limited security, scalability, and flexibility when managing user information and item records.**

Solution

Our proposed solution to this problem is:

A secure, scalable, and intelligent Lost & Found Management System that can automate item management, securely handle user access, and use AI-based matching techniques to identify potential matches between lost and found items

Objectives

- **The main objective of this project is to develop a secure, scalable, and intelligent Lost & Found Management System using SOA and Microservices Architecture.**
- **The system aims to provide secure user authentication and role-based authorization using Spring Security and JWT, enable dynamic service registration and discovery using Eureka, and manage service communication through Spring Cloud API Gateway.**
- **It also aims to integrate an AI Matching Service using TF-IDF and Cosine Similarity to automatically identify potential matches between lost and found item descriptions.**
- **The project further focuses on improving scalability, maintainability, modularity, and fault tolerance through independent microservices and databases, while following Agile development and testing practices.**

Literature Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work
Söylemez et al., 2022	Challenges and Solution Directions of Microservice Architectures: A Systematic Literature Review	•Systematic Literature Review of 3,842 papers; 85 selected as primary studies	•Identifies major challenges in Microservices Architecture, including scalability, reliability, availability, service management and maintainability.	•Further research is required to address the identified challenges and improve successful MSA adoption.
Tokmak et al., 2024	Web Service Discovery: Rationale, Challenges, and Solution Directions	•Systematic Literature Review; 54 papers analyzed	•Highlights service discovery as an important challenge in SOA and microservice environments and analyzes existing discovery techniques.	•More effective and scalable service discovery approaches are required as the number of services increases.
Assessment of	Assessment of Performance and its		•Shows that microservice performance and	•Existing studies use diverse approaches, mal

Problem Gap Identification

Problem Gap	Solution	Paper
Microservices introduce challenges in service management and maintainability.	Independent Spring Boot microservices for User, Item, AI Matching and Notification.	Söylemez et al. (2022)
Increasing number of services makes service discovery difficult.	Eureka-based dynamic service registration and discovery.	Tokmak et al. (2024)
Existing scalability research lacks standardized and comprehensive assessment approaches.	API Gateway, independent services, and database-per-service architecture.	Rodrigues et al. (2025)

Innovation ,Creativity and Novelty



AI-Based Item Matching: Uses TF-IDF and Cosine Similarity to automatically identify potential matches between lost and found item descriptions.



Microservices Architecture: Divides the system into independent services, making the application modular, scalable, and easier to maintain.



Secure Authentication: Uses Spring Security and JWT to provide secure authentication and role-based authorization.



Dynamic Service Discovery: Uses Eureka Server to automatically register and discover microservices without fixed service dependencies.



Centralized API Gateway: Uses Spring Cloud Gateway as a single entry point for request routing, security filtering, CORS, and load balancing.



Independent Database Architecture: Maintains separate databases for individual services, reducing data coupling and improving scalability.



Automated Notifications: Notifies users when the AI service identifies a potential match for their lost or found item.



Integrated Enterprise Solution: Combines AI, security, SOA, microservices, service discovery, and notifications into a single Lost & Found Management System.

Feasibility and Project Plan

Duration

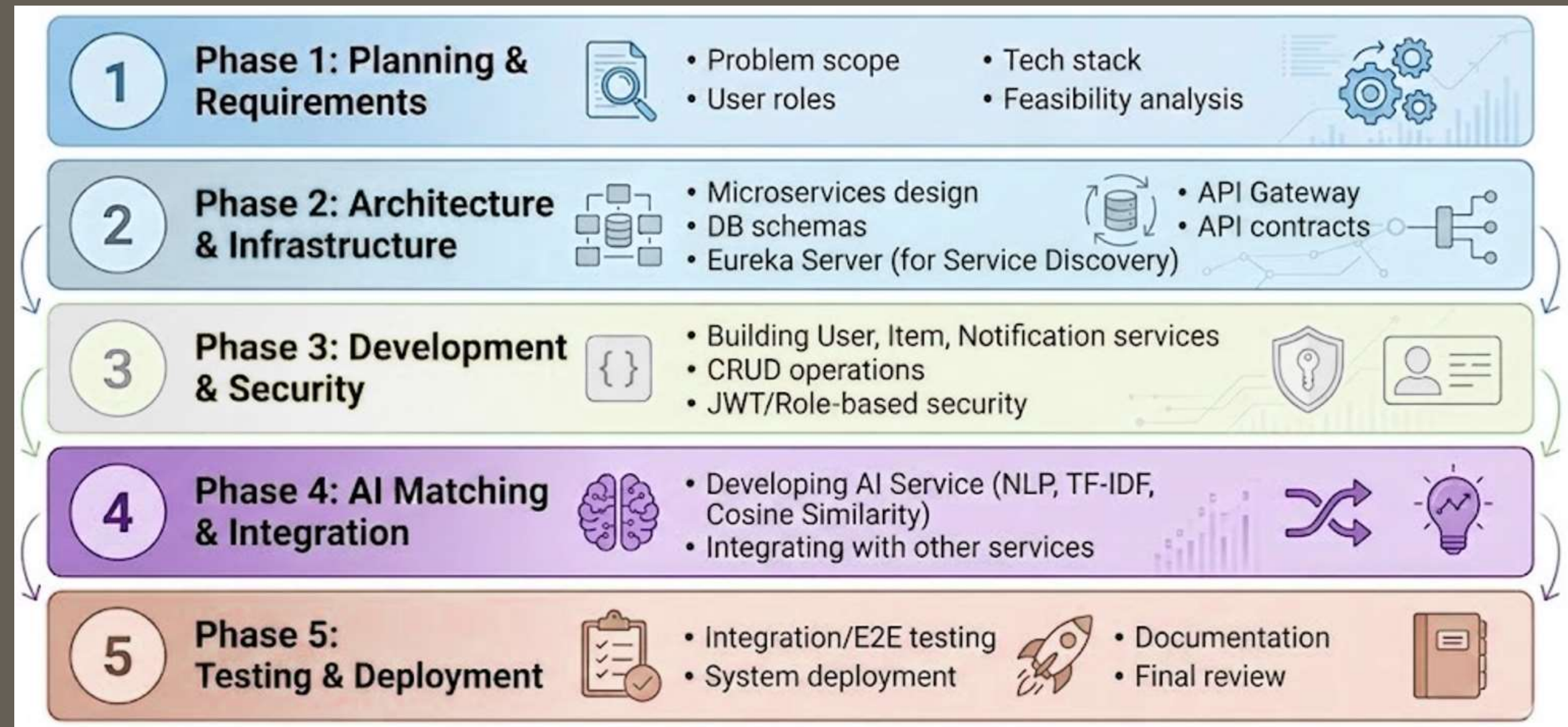
1 WEEKS

1-2 WEEKS

1 WEEK

1-2 WEEKS

1 WEEK



System Status

Environment	test
Data center	default

Current time	2026-09-24T12:21:23 +0530
Uptime	00:03
Lease expiration enabled	false
Renews threshold	10
Renews (last min)	2

DS Replicas

[localhost](#)

Instances currently registered with Eureka

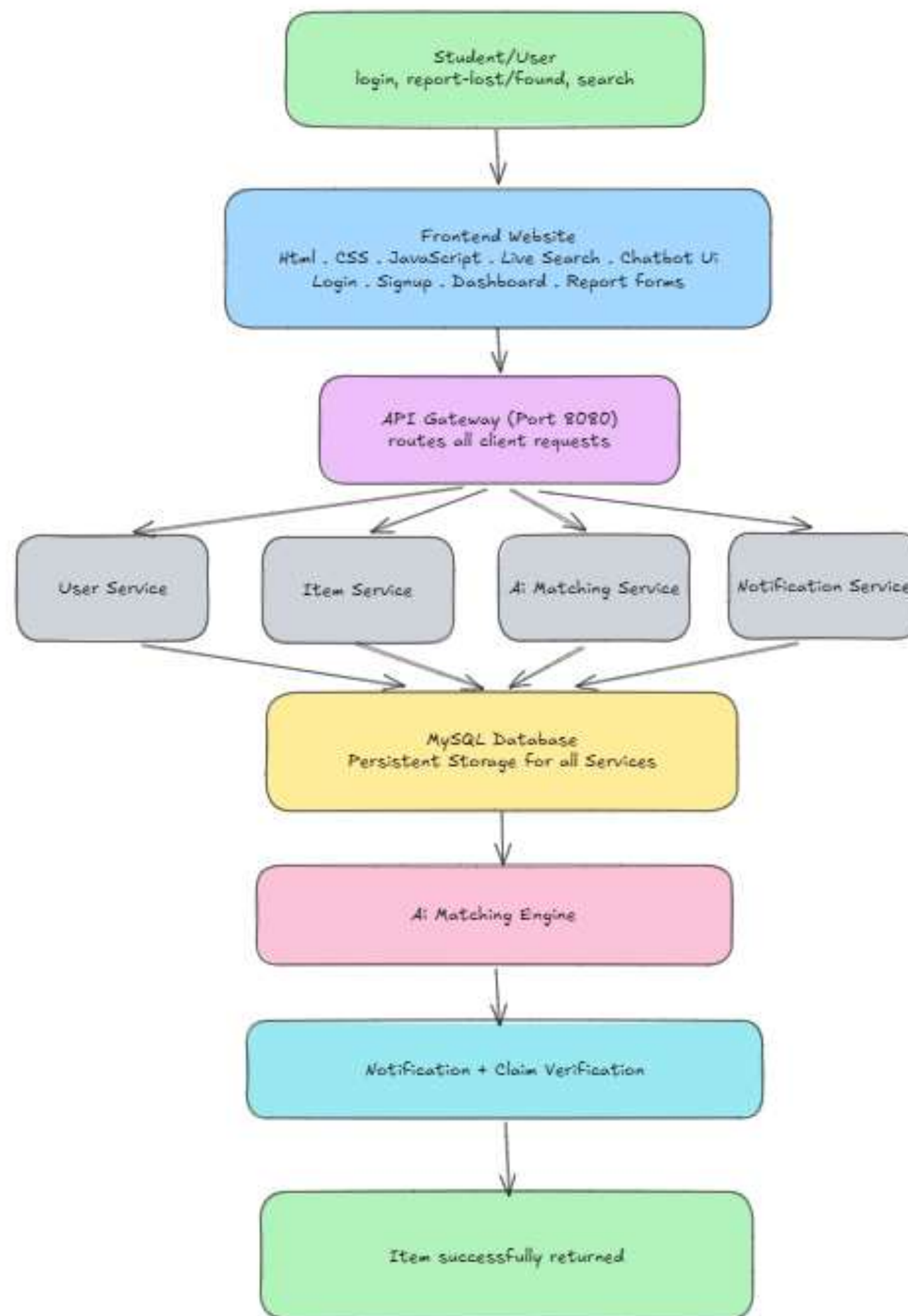
Application	AMIs	Availability Zones	Status
AI-MATCHING-SERVICE	n/a (1)	(1)	UP (1) - DESKTOP-ICCG05F:ai-matching-service:8084
API-GATEWAY	n/a (1)	(1)	UP (1) - DESKTOP-ICCG05F:api-gateway:8080
ITEM-SERVICE	n/a (1)	(1)	UP (1) - DESKTOP-ICCG05F:item-service:8082
NOTIFICATION-SERVICE	n/a (1)	(1)	UP (1) - DESKTOP-ICCG05F:notification-service:8083
USER-SERVICE	n/a (1)	(1)	UP (1) - DESKTOP-ICCG05F:USER-SERVICE:8081

General Info

Name	Value
total-avail-memory	114mb
num-of-cpus	8
current-memory-usage	86mb (75%)
server-uptime	00:03
registered-replicas	http://localhost:8761/eureka/
unavailable-replicas	http://localhost:8761/eureka/ ,
available-replicas	

Instance Info

Name	Value
ipAddr	10.250.21.150
status	UP



Data Sets:

<https://cocodataset.org>

https://storage.googleapis.com/openimages/web/download_v7.html

https://cvgl.stanford.edu/projects/lifted_struct/

https://www.vision.caltech.edu/datasets/cub_200_2011



Welcome Back

Campus Email (e.g. rollno@klh.edu.in)

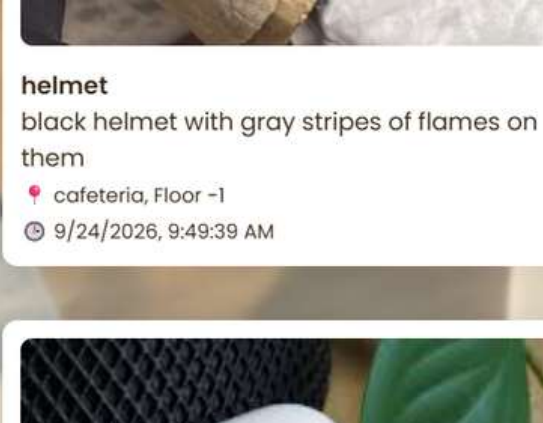
Password

Login

Don't have an account? [Sign up](#)



bottle
clear bottle with fruits print on them
📍 102, Floor 1
🕒 9/24/2026, 9:48:51 AM



helmet
black helmet with gray stripes of flames on them
📍 cafeteria, Floor -1
🕒 9/24/2026, 9:49:39 AM



airpods
white apple airpods v5
📍 library, Floor 2
🕒 9/24/2026, 9:48:10 AM



This page says

Verification link sent to 2420030697@klh.edu.in (simulation).

OK



KLH Lost & Found

[Admin](#) [+ Report Item](#) [Logout](#) [Timeline](#)

Search items or descriptions...

All Floors

All Types

Search
by
Image

Use
Camera

Camera

Create Account

Dama Poojitha

2420030697

2420030697@klh.edu.in

....

....

Sign up

Already have an account? [Login](#)



airpods
white apple airpods v5
📍 library (Floor 2)

[Chat](#) [AI Match](#)



bottle
clear bottle with fruits print on them
📍 102 (Floor 1)

[Chat](#) [AI Match](#)



helmet
black helmet with gray stripes of flames on them
📍 cafeteria (Floor -1)

[Chat](#) [AI Match](#)



laptop charger
hp z book laptop charger
📍 3rd floor 302 (Floor 3)

[Chat](#) [AI Match](#)

Conclusion

“The project provides a scalable and efficient microservices-based platform to simplify the process of reporting, finding, and recovering lost items on campus.”



Thank
You